EXP13: Minimax Algorithm for Gaming

Name: Y. Roshitha (192424400)

Aim:

To develop a Python program that implements the Minimax Algorithm to determine the optimal value for the Maximizer in a two-player game represented as a game tree.

Algorithm:

Step 1: Represent the game tree using the leaf node values (terminal utility values).

Step 2: Start the recursive minimax function from the root node with the Maximizer's turn.

Step 3: If the current node is a leaf node, return its value.

Step 4: Otherwise, if it is the Maximizer's turn, recursively evaluate all child nodes and return the maximum value.

Step 5: If it is the Minimizer's turn, recursively evaluate all child nodes and return the minimum value.

Step 6: Repeat Steps 3 to 5 for every node until the root node's value is computed.

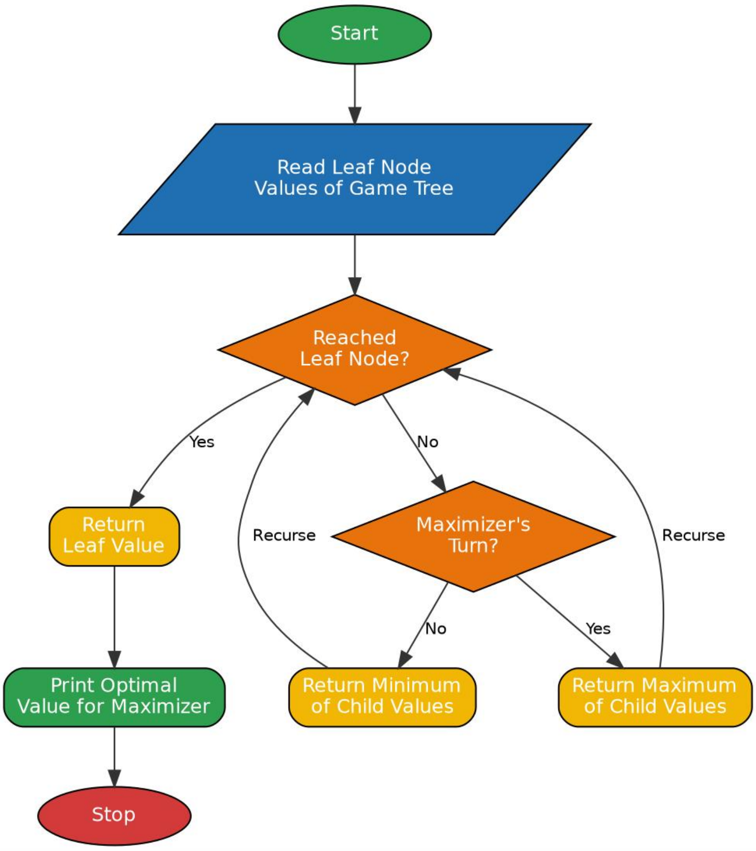
Step 7: Display the optimal value obtained for the Maximizer at the root node.

Source Code:

```
# Program to implement the Minimax Algorithm for Gaming

import math

def minimax(depth, node_index, is_max, scores, max_depth):
    if depth == max_depth:
        return scores[node_index]

    if is_max:
        best = -math.inf
        for i in range(2):
            value = minimax(depth + 1, node_index * 2 + i, False, scores, max_depth)
            best = max(best, value)
        return best
    else:
        best = math.inf
        for i in range(2):
            value = minimax(depth + 1, node_index * 2 + i, True, scores, max_depth)
            best = min(best, value)
        return best


scores = [3, 5, 2, 9, 12, 5, 23, 23]
max_depth = int(math.log2(len(scores)))

optimal_value = minimax(0, 0, True, scores, max_depth)
print("Leaf node values  :", scores)
```

```
print("Tree depth          :", max_depth)
print("Optimal value for the Maximizer:", optimal_value)
```

Flowchart:



Output:

```
>>>
==================== RESTART: C:/Users/yanar/Documents/CSA1717/EXP13.py ====================
Leaf node values  : [3, 5, 2, 9, 12, 5, 23, 23]
Tree depth        : 3
Optimal value for the Maximizer: 12
>>>
```

Result:

Thus the program to implement the Minimax Algorithm for gaming was written, executed successfully and the output was verified.